A sharp weak-driving phase-lag bound for a symmetric three-state Markov cycle

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

Periodically driven stochastic systems lag behind their driving protocol. A generalized phase lag has a sharp kinetic bound for two-state systems, while its extension to multistate networks is open. We treat the smallest complete cycle with a common kinetic prefactor and symmetric local-detailed-balance splitting. For arbitrary finite static offsets and arbitrary non-gauge finite Fourier driving, the leading weak-driving coefficients are positive modal sums. A strict Dirichlet-form estimate places both relaxation rates above twice the common prefactor. The resulting phase bound is stronger than the multistate conjectured bound, is strict at every finite offset, and has a sharp constant approached when one state is removed by an infinite-offset limit. The result is perturbative for each fixed input and does not resolve the finite-amplitude conjecture.

Keywords: stochastic thermodynamics; periodic Markov processes; linear response; phase lag; spectral gap

1 Introduction

Finite-state Markov models are a standard description of small driven systems in stochastic thermodynamics [1]. Under periodic driving, their long-time state is itself periodic but generally lags behind the instantaneous equilibrium state. Degünther, Koyuk and Seifert introduced a generalized phase that quantifies this lag for arbitrary periodic energy protocols [2]. For two states they proved an all-amplitude upper bound controlled by the kinetic prefactor. When the forward and backward rates split the energy difference symmetrically, their sharper bound contains an additional factor of one half. For general networks they proposed a weaker bound and reported numerical evidence, but did not prove it. A later dissertation retains the arbitrary-state proof as an open task [3].

This note isolates the complete three-state graph with a common kinetic prefactor and symmetric splitting. The full finite-amplitude question remains open even in this restricted model. We instead prove an exact nearby result: at the leading nonzero order in the driving amplitude, every non-gauge finite Fourier protocol obeys the stronger factor-two inequality associated with the symmetric two-state problem. The mechanism is spectral. Linear response expresses the relevant observables as positive sums over the two relaxation modes, while a three-point Dirichlet identity places both relaxation rates strictly above twice the common prefactor. The constant is uniform and sharp as an infimum in an effective-two-state limit.

Section 2 fixes the physical conventions and states the result. Section 3 derives the modal response formula, section 4 proves the relaxation bound and its sharpness, and section 5 records numerical convention checks and the limitations of the conclusion.

2 Model, observables, and result

Consider the complete graph on three states. The symbol k_{ij} denotes the departure rate from state i to state j, and the column probability vector obeys

$$\dot{p}_i(t) = -\sum_{j\neq i} \left[p_i(t)k_{ij}(t) - p_j(t)k_{ji}(t)\right]. \tag{1}$$

We set $k_{\mathrm{B}}T = 1$ and use the symmetric local-detailed-balance rates

$$k_{ij}(t) = k_0 \exp\!\left[\frac{E_i(t) - E_j(t)}{2}\right], \qquad k_0 > 0. \tag{2}$$

Thus $k_{ij}/k_{ji} = \exp(E_i - E_j)$. All rates are finite and strictly positive in the domain considered below.

Let $T = 2\pi/\Omega$, with $\Omega > 0$, and write

$$E_i(t) = E_i^0 + \varepsilon e_i(t), \qquad e_i(t) = \sum_{n=1}^{M} \left[c_{ni}\sin(n\Omega t) + d_{ni}\cos(n\Omega t)\right], \tag{3}$$

where M and all real offsets and coefficients are finite. The positive, normalized T-periodic state is unique because the time-dependent generator is irreducible. Define the two period-averaged observables

$$\mathcal{A} = \frac{1}{T}\int_0^T p(t)^{\mathsf{T}}\ddot{E}(t)\,dt, \qquad \mathcal{B} = \frac{1}{T}\int_0^T p(t)^{\mathsf{T}}\dot{E}(t)\,dt. \tag{4}$$

Their dimensions are time^{-2} and time^{-1}, respectively. Whenever both are positive, the generalized phase of Ref. [2] is

$$\Delta\varphi = \arctan\!\left(\frac{\Omega\mathcal{B}}{\mathcal{A}}\right). \tag{5}$$

A common time-dependent shift of all three energies changes neither the rates nor these period averages. A drive is called non-gauge when it is not entirely of this form.

Result 1 (strict weak-driving bound and sharpness)**.** *For every fixed finite set of offsets, every $k_0, \Omega > 0$, and every non-gauge finite real Fourier drive in Eq.* (3)*, the periodic-state observables have the fixed-input expansions*

$$\mathcal{A}(\varepsilon) = \varepsilon^2 A_2 + O(\varepsilon^3), \qquad \mathcal{B}(\varepsilon) = \varepsilon^2 B_2 + O(\varepsilon^3), \tag{6}$$

with

$$B_2 > 0, \qquad A_2 > 2k_0 B_2. \tag{7}$$

Consequently, $\mathcal{A}$ and $\mathcal{B}$ are positive for sufficiently small positive ε, and

$$\lim_{\varepsilon\to 0^+} \Delta\varphi < \arctan\!\left(\frac{\Omega}{2k_0}\right). \tag{8}$$

The uniform constant 2 is sharp: the infimum of $A_2/(k_0B_2)$ over the stated finite domain is 2, although equality is attained only in an excluded infinite-offset limit.

The remainder in Eq. (6) is local for each fixed choice of offsets, Fourier coefficients, harmonic cutoff, k_0, and Ω. No uniform radius in these inputs is asserted.

3 Reversible modal response

At $\varepsilon = 0$, the equilibrium distribution is

$$\pi_i = \frac{e^{-E_i^0}}{Z}, \qquad Z = \sum_{j=1}^{3} e^{-E_j^0}. \tag{9}$$

Put $D = \mathrm{diag}(\pi)$ and $s = \sqrt{\pi}$ componentwise. If L_0 is the column generator obtained from Eq. (1), detailed balance makes

$$K = -D^{-1/2} L_0 D^{1/2} \tag{10}$$

real symmetric and positive semidefinite. Its kernel is spanned by s, and it has two positive relaxation eigenvalues λ_a on $s^\perp$.

Remove the common-energy component of the perturbation by defining

$$v(t) = D^{1/2}\left[e(t) - (\pi^{\mathsf{T}} e(t))\mathbf{1}\right], \qquad s^{\mathsf{T}} v(t) = 0. \tag{11}$$

The first variation of the instantaneous equilibrium distribution is $-D^{1/2}v$. Writing the periodic state as

$$p(t) = \pi + \varepsilon D^{1/2} z(t) + O(\varepsilon^2), \qquad s^{\mathsf{T}} z(t) = 0, \tag{12}$$

and differentiating $L(E)\pi(E) = 0$ gives the forced relaxation equation

$$\dot{z} = -K(z + v). \tag{13}$$

Let u_1, u_2 be orthonormal eigenvectors of K on $s^\perp$. With $\omega_n = n\Omega$, expand

$$v(t) = \mathrm{Re} \sum_{n=1}^{M} \sum_{a=1}^{2} c_{an} u_a e^{i\omega_n t}. \tag{14}$$

Equation (13) then has the unique periodic solution

$$z(t) = \mathrm{Re} \sum_{n=1}^{M} \sum_{a=1}^{2} \left(-\frac{\lambda_a}{\lambda_a + i\omega_n}\right) c_{an} u_a e^{i\omega_n t}. \tag{15}$$

The equilibrium contribution to each observable in Eq. (4) is a periodic total derivative. Moreover, normalization gives $(D^{1/2}z)^{\mathsf{T}}\dot{e} = z^{\mathsf{T}}\dot{v}$, and likewise for the second derivative. Orthogonality of modes and harmonics therefore yields

$$B_2 = \frac{1}{2} \sum_{a=1}^{2} \sum_{n=1}^{M} |c_{an}|^2 \frac{\lambda_a \omega_n^2}{\lambda_a^2 + \omega_n^2}, \tag{16}$$

$$A_2 = \frac{1}{2} \sum_{a=1}^{2} \sum_{n=1}^{M} |c_{an}|^2 \frac{\lambda_a^2 \omega_n^2}{\lambda_a^2 + \omega_n^2}. \tag{17}$$

A non-gauge drive has at least one nonzero c_{an}, so $B_2 > 0$. More importantly, A_2/B_2 is a positive weighted average of the excited relaxation rates. It remains only to bound those two rates.

For completeness, the fixed-input remainder follows without a large-time asymptotic assumption. The finite-dimensional generator depends analytically on ε. Its one-period propagator is a positive stochastic matrix for finite rates, so its normalized unit-eigenvalue fixed point is simple. The periodic state and the observables are therefore analytic near $\varepsilon = 0$. Together with the explicit vanishing of the lower orders, this gives Eq. (6).

4 Strict relaxation bound and sharpness

Write

$$r_i = e^{-E_i^0/2} > 0, \qquad Z = \sum_{i=1}^{3} r_i^2, \qquad \pi_i = \frac{r_i^2}{Z}. \tag{18}$$

The backward generator associated with Eq. (1) is reversible with respect to π and has the same two nonzero decay rates as K. For a real state observable f, its Dirichlet form is

$$\mathcal{E}(f,f) = \frac{k_0}{Z} \sum_{i<j} r_i r_j (f_i - f_j)^2, \tag{19}$$

whereas the exact pairwise representation of the equilibrium variance is

$$\mathrm{Var}_\pi(f) = \frac{1}{Z^2} \sum_{i<j} r_i^2 r_j^2 (f_i - f_j)^2. \tag{20}$$

For a pair i, j, let k be the remaining state. Then

$$Z - 2r_i r_j = (r_i - r_j)^2 + r_k^2 > 0. \tag{21}$$

Subtracting twice k_0 times Eq. (20) from Eq. (19) gives the identity

$$\mathcal{E}(f,f) - 2k_0 \,\mathrm{Var}_\pi(f) = \frac{k_0}{Z^2} \sum_{i<j} r_i r_j \left[(r_i - r_j)^2 + r_k^2\right] (f_i - f_j)^2. \tag{22}$$

It is strictly positive for every nonconstant f. The Rayleigh principle therefore places both nonzero relaxation rates strictly above $2k_0$:

$$\lambda_1 > 2k_0, \qquad \lambda_2 > 2k_0. \tag{23}$$

Substitution into Eqs. (16)–(17) proves Eq. (7) term by term.

To show sharpness, take

$$E_1^0 = E_2^0 = 0, \qquad E_3^0 = H, \tag{24}$$

and drive only the antisymmetric combination $e_1 = -e_2$, $e_3 = 0$. The excited mode is exactly antisymmetric and has

$$\frac{\lambda_{\mathrm{as}}}{k_0} = 2 + e^{-H/2}. \tag{25}$$

Only this relaxation mode contributes, so $A_2/(k_0 B_2) = 2 + e^{-H/2}$ independently of the driving frequency. Every finite H remains inside the theorem and gives strict inequality; sending $H \to \infty$ removes the third state and approaches the two-state constant. Figure 1 displays this exact family.

5 Checks, interpretation, and limitations

The proof above is exact. Numerical calculations were used only to detect convention errors and to verify that the full nonlinear periodic solution approaches the modal formula. One implementation evolves all three probabilities through a full-period propagator. A second eliminates normalization and solves the remaining periodic equations by Fourier collocation. Table 1 gives two fixed examples. Both use $k_0 = 1$. The last column reports the larger of the absolute differences in $\mathcal{A}$ and $k_0\mathcal{B}$ between the two representations at $\varepsilon = 0.05$; it is not an error bound for the analytic theorem.

The generic case uses offsets $(-0.8, 0.3, 1.1)$ and two sine/cosine harmonics; the second uses offsets $(0, 0, 7)$ and an antisymmetric single-harmonic drive. In both cases all probabilities remain

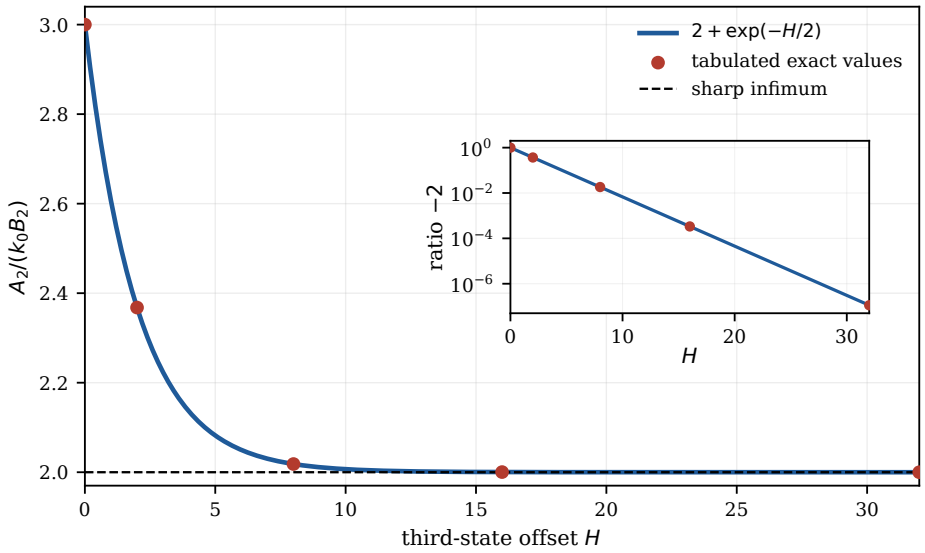


Figure 1: Exact sharpness family in Eq. (25). The solid curve and selected markers are generated from the analytic formula, not from a fit. The inset shows the positive margin above 2 on a logarithmic scale. The dashed line is the uniform infimum and is reached only at the excluded $H = \infty$ limit.

Table 1: Convention and perturbative checks. The linear-response (LR) ratio is computed from Eqs. (16)–(17); the nonlinear ratio uses the full periodic state at $\varepsilon = 0.025$.

case	Ω/k_0	LR $A_2/(k_0B_2)$	nonlinear $\mathcal{A}/(k_0\mathcal{B})$	max. difference
generic two-harmonic	1.3	3.3034360094	3.3024363997	1.47×10^{-15}
offset family, $H = 7$	0.8	2.0301973834	2.0302467618	1.22×10^{-17}

positive, normalization and periodic closure are at roundoff, and the scaled nonlinear observables converge to the separately evaluated coefficients as ε decreases. The supplied scripts and captured outputs specify all coefficients, tolerances, and collocation sizes.

Physically, Eq. (22) shows why merely adding the smallest cycle does not produce excessive lag under fully symmetric kinetics in linear response. The conductance weight $r_i r_j/Z$ of every edge dominates twice its contribution to the equilibrium variance, with a strictly positive contribution from the third state. Each relaxation channel is therefore faster than the symmetric two-state scale, and the phase inherits the stronger bound through a positive modal average.

This conclusion has several precise limits. It is a leading-order result as $\varepsilon \to 0$ for fixed finite input. It does not provide a uniform amplitude radius and does not prove either the stronger or weaker inequality at arbitrary finite amplitude. It also does not cover missing edges, asymmetric rate splitting, heterogeneous edge prefactors, transient rather than periodic states, singular offsets, more than three states, or a drive with unbounded harmonic content. Thus the original all-amplitude complete-three-state question remains unresolved. Any violation there would have to be a genuinely nonlinear-amplitude effect, not a failure already present in this symmetric linear-response mechanism.

6 Conclusion

For the complete symmetric three-state Markov model, the leading weak-driving phase is controlled by a positive average of two reversible relaxation rates. A strict three-point Dirichlet identity places each rate above $2k_0$, proving the factor-two phase-lag bound for arbitrary finite

offsets and arbitrary non-gauge finite Fourier driving. The constant is sharp only through an effective-two-state infinite-offset limit. The result settles a coherent perturbative subdomain but leaves the source's finite-amplitude multistate conjecture open.

Code and data availability

No external data set is used. The reproducibility package contains the exact figure script and tabulated values, a full-period three-coordinate propagator, an independent two-coordinate Fourier-collocation solver, and a modal-response checker. The archived environment is Python 3.12.14 with NumPy 2.3.5, SciPy 1.18.1, and Matplotlib 3.11.2. The accompanying README gives regeneration commands and distinguishes numerical checks from the analytic proof.

Acknowledgements and disclosure of AI assistance

OpenAI Codex using model `gpt-5.6-sol` assisted with problem scoping, analytic derivations, code generation and execution, numerical verification, literature discovery, and preparation of this draft. A separately initialized session of the same model reconstructed and stress-tested the central calculation. The AI system is not an author. Before any submission, the named author must critically review the complete manuscript, independently confirm the claims and references, and update this disclosure as required by the target journal.

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